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IDS Pulse Overview

Why IDS Pulse?

- Provides real-time reporting from the assembly floor directly to customers.
- Bridges the gap between Quality Representatives, Managers, and Clients.
- Eliminates delayed notifications, manual sheets, and unverified data.
- Ensures operations are visible, traceable, and instantly actionable.

Core Software Architecture

- Mobile representative App: For field reps to log quality issues on the floor.
- Web Customer Portal: For clients to track alerts, hours, and approve shifts.
- Web Admin Dashboard: For corporate managers to control rates and billing.
- Real-Time Cloud Sync: Auto-saves every entry to the database instantly.

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Mobile Representative App Flow

Shift Startup

- Representative logs in securely.
- Selects the plant location.
- Clocks in to open a new shift.
- Downloads active daily tasks checklist.

Shift Active

Floor Activities

- Scans parts barcodes or QR codes.
- Captures defect photos with notes.
- Draws directly on defect photos.
- Logs rework details and expenses.

Shift End

Shift Walk & Summary

- Performs physical walkthroughs.
- Completes 4-area walk checklist.
- Reviews reports for accuracy.
- Submits summary to customer.

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Incident Report Wizard (On-Floor Logging)

1 Capture & Mark

Rep captures up to three photos (wide, medium, close-up) and draws red annotations on the screen to highlight the defect.



2 Scan Part & Bin

Rep scans the part barcode and QR bin tag. System auto-fills part descriptions, supplier names, and details from the database.



3 Describe Defect

Rep selects standard defects, writes brief concern details, and categorizes the concern (PRR or QR) for client notification.



4 AI Check & Send

App checks for recent duplicate logs using Jaccard similarity. Rep previews the HTML email and clicks submit to notify the supplier.

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Floor Security & Incident History features

Shift Handover Protection

- Detects if another representative had an active shift on the same device.
- Auto-locks the previous draft shift report to prevent tampering.
- Presents handover alert requiring validation before continuing.

Offline Capability & History

- Offline Mode: saves reports to LocalStorage if internet drops on floor.
- Syncs background data directly to Supabase once back online.
- Incident History tab lets representatives review past reports directly.

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Web Corporate Dashboard (For Administrators)

Operations Management

- Incident Defective Feed: Real-time feed of logged defect activities on the floor.
- Visual Defect Heatmap: Interactive part diagram showing defect clusters over time.
- Daily Tasks Planner: Dispatches specific floor tasks to representatives.
- Shift Walkthrough Summaries: Log of walkthrough checklists and daily tasks.

Finance & Billing Control

- Timesheets Payroll Panel: Records of logged hours, expenses, and rep mileage.
- Client Invoice Generator: Auto-calculates invoice details by client.
- Overtime Approval Queue: Reviews escalated extra hours requests.
- Expense Approval Portal: Reviews filed claims and captures receipts.

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Web Customer Portal (For Clients)

Live Floor Progress

- Track QRE representative shift status.
- Monitor active hours in real-time.
- View progress bar of weekly hours.
- Review plant coverage status.

Overtime Review Queue

- View extra hours requests from reps.
- Approve/Reject overtime with comments.
- Approved items move to admin queue.
- Transparent, timestamped audit trail.

Published Shift Reports

- Access shift summaries published by reps.
- Audit hours billed to your supplier.
- Compare actual vs budgeted coverage.
- Rates are hidden from customer view.

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Overtime (Extra Hours) Approval Workflow



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Expense Claims Approval Workflow

